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Pages: 2

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APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

B.Tech Degree S3 (S,FE) / S1 (PT)(S) June 2024 (2019 Scheme)/S3 (WP)(R) December 2023 Examination



Course Code: MAT201

Course Name: Partial Differential Equations and Complex Analysis

Max. Marks: 100

Duration: 3 Hours

PART A

Answer all questions. Each question carries 3 marks

Marks

- 1 Form the partial differential equation by eliminating arbitrary function from the relation $z = x - y + f(x^2 + y^2)$. (3)
- 2 Solve by direct integration: $\frac{\partial^2 z}{\partial x \partial y} = \sin(3x + 4y)$. (3)
- 3 Write all possible solutions of one-dimensional wave equation. (3)
- 4 Write any three assumptions used in the derivation of one-dimensional heat equation. (3)
- 5 Check whether $f(z) = z - \bar{z}$ is analytic or not. (3)
- 6 Test the continuity of $f(z)$ at the point $z = 0$, where $f(z) = \begin{cases} \frac{\text{Im } z}{|z|}, & z \neq 0 \\ 0, & z = 0 \end{cases}$ (3)
- 7 Find the value of the integral $\int_C z \, dz$ where C is the part of the unit circle from $-i$ to i in the right half plane. (3)
- 8 Evaluate $\oint_C \frac{\cos z}{z^2 + 9} dz$, where C is the circle $|z| = 1$. (3)
- 9 Find the Laurent series expansion of $\frac{1}{1-z^2}$ about $z=0$ (3)
- 10 Find the singularities and nature of singularities of the function $f(z) = z^2 \cos \frac{1}{z}$ (3)

PART B

Answer any one full question from each module. Each question carries 14 marks

Module 1

- 11(a) Solve the PDE $x(y^2 - z^2)p + y(z^2 - x^2)q = z(x^2 - y^2)$. (7)
- (b) Solve by Method of Separation of variable $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial t} = 0, u(x, 0) = 4e^{-3x}$. (7)
- 12(a) Solve $2xz - px^2 - 2qxy + pq = 0$. (7)
- (b) Solve $(y^2 + z^2)p - xyq = xz$ (7)

08000MAT201122301

Module 2

- 13(a) A tightly stretched string with fixed end points, on $x = 0$ and $x = l$ is initially at rest in its equilibrium position. If it is vibrating by giving each of its points a velocity $\lambda x(l - x)$. Find the displacement of the string at any distance x from one end at any time t . (7)
- (b) Derive One Dimensional Wave equation. (7)
- 14(a) Derive all possible solutions of one-dimensional wave equation. (7)
- (b) Find the temperature $u(x, t)$ in a homogeneous rod of length $2m$, whose ends are kept at 0°C and whose initial temperature is given by $f(x) = 100(2x - x^2)$, $0 \leq x \leq 2$ (7)

Module 3

- 15(a) Show that $f(z) = e^z$ is analytic for all z in the complex plane and find its derivative. (7)
- (b) Find image of the circle $|z - \frac{1}{2}| \leq \frac{1}{2}$ under the transformation $w = \frac{1}{z}$. (7)
- 16(a) Show that the function $u = \cos x \cosh y$ is harmonic and find its harmonic conjugate. (7)
- (b) Show that an analytic function $f(z) = u + iv$ is constant if $|f(z)|$ is constant (7)

Module 4

- 17(a) Evaluate $\int_C \frac{2z+1}{z^2-2z} dz$ where $|z| = 1$. (7)
- (b) Find the Taylor series expansion of $f(z) = \sin 2z$ about $z = \frac{\pi}{2}$. (7)
- 18(a) Evaluate $\int_C |z| dz$ where C is the part of the circle $|z| = 3$ from 3 to -3 in the upper half plane. (7)
- (b) Evaluate $\int_C \frac{e^{3z}}{(z+1)^4} dz$ where C is the circle $|z - 1| = 3$. (7)

Module 5

- 19(a) Find Laurent series expansion $f(z) = \frac{-2z+3}{(z^2-3z+2)}$ valid in (7)
- i. $1 \leq |z| \leq 2$
- ii. $|z| \geq 2$
- (b) Evaluate $\oint_C \frac{e^z}{z^2(z+1)} dz$ over the circle $|z| = 2$ (7)
- 20(a) Evaluate $\int_C \tan z dz$ where C is circle $|z| = 2$. (7)
- (b) Evaluate $\int_0^{2\pi} \frac{d\theta}{\sqrt{2}-\cos \theta}$ (7)